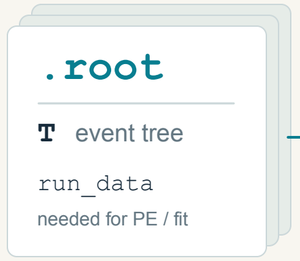
Analysis

ROOT → selected signals → views & deconvolution



Read one event

Event fields

rate | ev.A | ev.W2

hit[]

All recorded hits stay inside their event

geometry/load_hits.C

```
264  tree->GetEntry(event)
266  const int count = detector->GetNdata();
```

hit.det

joins hit records to GDML detector faces.

PE rates across files

Check run identity, histories and current.
Combine compatible runs; normalize once.

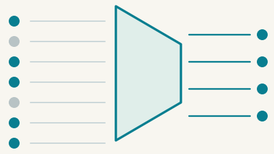
geometry/run.py:549-561
root/Deconvolution.C:320-329

Two analysis paths

Explore hits

Editable display cuts

1 Select



Detector ID
Particle / K
Direction

geometry/load_hits.C

```
282  selection.EvalInstance(scalar ? 0 : hit)
285  if (keep == 0) continue;
```

2 Assign a contribution

Hits	1 per selected record
PE	Expected PE × event rate
Current	PE/s × electron charge

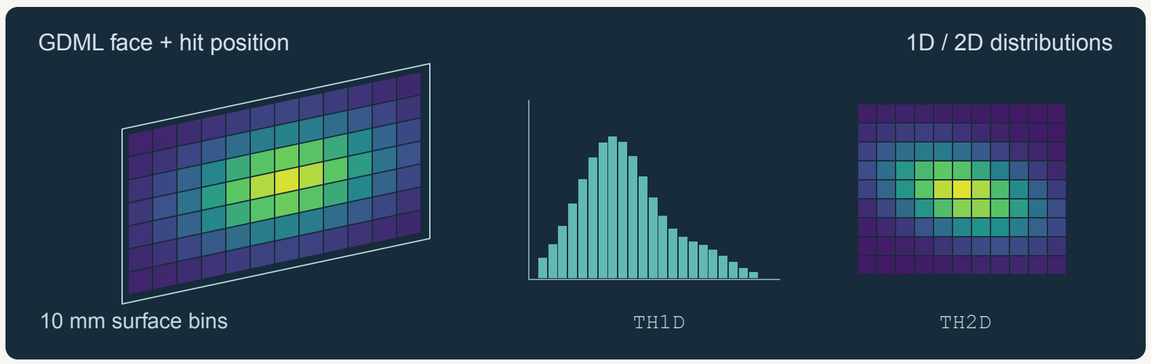
Shared PE models

Main: position / angle fit × energy response

ShowerMax: module-local lookup

app/responses.py · root/pe_response.hh
vendor/portable/scan.C

3 Bin the same selected signal



geometry/load_hits.C

```
49   hist.Fill(value,weight);
335  bins[{face,int(std::floor(bu/10)),int(std::floor(bv/10))}]+=weight;
```

Detector ID → face → local u, v → bin → color

The browser draws bins; it does not redo the ROOT selection.

Deconvolution

Separate fixed selection · PE weighted

1 Sum the selected PE in each tile

Check generators · detector coverage · run settings · PE support

One event · illustrative PE values

2 3 → a = 5 PE · Tile 1

1 → b = 1 PE · Tile 2

Multiply tile totals; rate once

$$w \times \begin{bmatrix} 5\times5 & 5\times1 \\ 1\times5 & 1\times1 \end{bmatrix}$$

w: event rate after file / current normalization [s^{-1}]

Video: hit counts. Here: summed expected PE.

In code: a, b are two entries of q.

root/Deconvolution.C

```
224  q[i]+=main->pe(h,k[j]);
```

2 Add the contributions from every event

Tile rates [PE/s]

$$R_1 = \Sigma\, w\, a$$

$$R_2 = \Sigma\, w\, b$$

Accumulated second moment B [PE^2/s]

$$B = \begin{bmatrix} \Sigma\, w\, a^2 & \Sigma\, w\, a\, b \\ \Sigma\, w\, b\, a & \Sigma\, w\, b^2 \end{bmatrix}$$

Σ sums over events; a, b and w belong to each event.
Repeat for all tile and ShowerMax pairs to fill B.

root/Deconvolution.C

```
232  double yi=*rate*q[i];
237  if(i<n&&j<n)out.noise[i*n+j]+=*rate*q[i]*q[j];
```

3 Solve the weighted system

$$C_{rs} = B_{rs} / (P^2\, T\, R_r\, R_s) \times 10^{18}$$

H = dilution × asymmetry adjustment

P: polarization · T: seconds · C: ppb^2

root/root_solver.hh:64-134

$$M = H^T C^{-1} H$$

σ^2 = diagonal of M^{-1}

Finite MC uncertainty propagates through H.

Cholesky + SVD

ROOT calculates → Compact JSON → Browser draws

Maps & distributions / matrices & detector correlations

geometry/view.js · hit-map-3d.js

geometry/deconvolution_bridge.C → deconvolution.js

Diagrams are schematic · Source excerpts are literal